

Shared-Lane Gram Ellipsoids and Exact Joint Covariance Geometry

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Abstract

A source-index decomposition of the finite V59 twin-prime Gate-B scalar exposes, inside each fixed output block, several probe vectors paired with one shared physical output lane. Independent marginal disks therefore do not describe the source-valid joint family. We determine that family exactly. For the analysis operator $V^*W = (\langle v_i, W \rangle)_i$ and Gram matrix $G = V^*V$, the radius- ρ ball maps onto

$$\{y \in \text{ran } G : y^* G^\dagger y \leq \rho^2\}.$$

We classify the exact-sphere image by the orthogonal slack $\ker V^*$, give the entrywise-conjugate physical covariance form, and distinguish independent group budgets from one global Hilbert budget. Repeated probes collapse two unit-disk marginals to a diagonal disk, while orthogonal probes give a Euclidean ball rather than a bidisk. These are exact structural results; they provide no arithmetic cancellation, strict 1/400 payment, full Gate B, or twin-prime conclusion.

1 The source-native joint object

TPC-247 writes the physical scalar as a finite source covariance and, for a declared disjoint hard partition, obtains

$$A_{cb} = P_c A_x P_b, \quad \beta_b = P_b \beta, \quad w_c = P_c w, \quad C_x = \sum_{c,b} \langle w_c, A_{cb} \beta_b \rangle.$$

For fixed c , set

$$v_{cb} := A_{cb} \beta_b \in P_c \mathcal{H}_x.$$

The vectors $(v_{cb})_b$ are all observed by the same w_c ; no independently variable lane w_{cb} appears in the physical source. Tagged direct-sum copies reproduce the scalar but repeat w_c once for every b , which is why their Cartesian geometry is not automatically source-valid [1, 2].

We therefore solve the following finite-dimensional problem before applying any weighted reassembly. Throughout, complex Hilbert inner products are conjugate-linear in the first slot.

2 Exact analysis-map ellipsoid

Let $v_1, \dots, v_m$ be ordered vectors in a finite-dimensional complex Hilbert space $\mathcal{H}$. Define

$$V : \mathbb{C}^m \longrightarrow \mathcal{H}, \quad Va = \sum_{i=1}^m a_i v_i, \quad G = V^*V,$$

so $G_{ij} = \langle v_i, v_j \rangle$. The Moore–Penrose inverse is denoted by $G^\dagger$.

Theorem 2.1 (ball image and minimum preimage). *For every $\rho \geq 0$,*

$$V^*\{W \in \mathcal{H} : \|W\| \leq \rho\} = \{y \in \text{ran } G : y^* G^\dagger y \leq \rho^2\}.$$

For each $y \in \text{ran } G$, the vector

$$W_0 = V G^\dagger y$$

is the unique minimum-norm preimage and

$$\min_{V^*W=y} \|W\|^2 = \|W_0\|^2 = y^* G^\dagger y.$$

Proof. In finite dimension, $\text{ran } V^* = \text{ran}(V^*V) = \text{ran } G$. If $y \in \text{ran } G$, the Moore–Penrose identities give

$$V^*W_0 = G G^\dagger y = y.$$

Furthermore,

$$\begin{aligned} \|W_0\|^2 &= \langle V G^\dagger y, V G^\dagger y \rangle \\ &= (G^\dagger y)^* G (G^\dagger y) = y^* G^\dagger y. \end{aligned}$$

Here we used that G and $G^\dagger$ are Hermitian and that $G G^\dagger y = y$. Every other preimage is uniquely $W_0 + k$ with $k \in \ker V^*$. Since $W_0 \in \text{ran } V = (\ker V^*)^\perp$,

$$\|W_0 + k\|^2 = \|W_0\|^2 + \|k\|^2.$$

The minimum formula and both inclusions in the ball identity follow. The same argument includes rank-zero G and $\rho = 0$. $\square$

The range condition is essential when the probes are linearly dependent: the quadratic inequality by itself must not be interpreted on all of $\mathbb{C}^m$.

3 Exact spheres and physical orientation

Theorem 3.1 (orthogonal-slack dichotomy). *Let $S_\rho = \{W : \|W\| = \rho\}$. If $\ker V^* \neq \{0\}$, then*

$$V^*S_\rho = \{y \in \text{ran } G : y^* G^\dagger y \leq \rho^2\}.$$

If $\ker V^ = \{0\}$, then*

$$V^*S_\rho = \{y \in \text{ran } G : y^* G^\dagger y = \rho^2\}.$$

Both formulas remain valid for $\rho = 0$.

Proof. For a point of minimum energy $e = y^* G^\dagger y \leq \rho^2$, choose a unit $k \in \ker V^*$ and replace W_0 by $W_0 + \sqrt{\rho^2 - e} k$. This reaches the sphere without changing y . If the kernel is zero, W_0 is the only preimage, so sphere membership is equivalent to $e = \rho^2$. $\square$

The V59 physical scalar uses the opposite orientation

$$z_i = \langle W, v_i \rangle = \overline{\langle v_i, W \rangle} = \overline{y_i}.$$

Corollary 3.2 (physical covariance ellipsoid). *The physical ball image is*

$$\{z \in \text{ran } \overline{G} : z^* (\overline{G})^\dagger z \leq \rho^2\}.$$

The sphere classification is Theorem 3.1 with G replaced by $\overline{G}$.

Proof. Entrywise conjugation carries $\text{ran } G$ onto $\text{ran } \overline{G}$, commutes with the Moore–Penrose inverse, and conjugates the real nonnegative energy $y^* G^\dagger y$ to the displayed expression. $\square$

4 Grouped budgets are not interchangeable

For output groups c , let $V_c^* W_c = y_c$ and $G_c = V_c^* V_c$.

Theorem 4.1 (product and global budgets). *For explicitly independent constraints $\|W_c\| \leq \rho_c$, the joint image is the Cartesian product*

$$\prod_c \{y_c \in \text{ran } G_c : y_c^* G_c^\dagger y_c \leq \rho_c^2\}.$$

For one global direct-sum constraint $\sum_c \|W_c\|^2 \leq \rho^2$, the exact image is instead

$$\left\{ (y_c)_c : y_c \in \text{ran } G_c, \quad \sum_c y_c^* G_c^\dagger y_c \leq \rho^2 \right\}.$$

Proof. The first domain is a Cartesian product and the map acts coordinatewise. For the second domain, Theorem 2.1 supplies the minimum squared norm for each group; direct-sum orthogonality adds these minima. Independent minimum preimages realize every point satisfying the summed inequality. $\square$

Thus “local ellipsoid” does not by itself determine a cross-group feasible set. The budget domain is part of the theorem.

5 Sharp adversarial fixtures

First take $v_1 = v_2 = e_1$ and $\rho = 1$. Then

$$G = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad G^\dagger = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

and the exact image is

$$\{(t, t) : |t| \leq 1\}.$$

Each coordinate marginal is the unit disk, but $(1, 0)$ is not jointly feasible. This refutes the bidisk promotion at rank one.

Second take $v_1 = e_1, v_2 = e_2$. The image is $\{(y_1, y_2) : |y_1|^2 + |y_2|^2 \leq 1\}$, again strictly smaller than the bidisk. If the ambient space contains an unused e_3 , the exact unit sphere maps onto that whole solid ball; without e_3 , it maps only onto its boundary.

Finally, two scalar groups with a global unit budget have local unit-disk marginals, but $(1, 1)$ has minimum global energy 2 and is forbidden. These examples isolate three different errors: ignoring Gram range, ignoring joint energy, and ignoring orthogonal sphere slack.

6 Certificate and route boundary

The exact certificate freezes a rank-one Gram pair, two full-rank Gram pairs, the sphere dichotomy, a global-budget obstruction, a zero-radius boundary, and a Gaussian-rational orientation fixture. An independent implementation reconstructs the Moore–Penrose identities and rejects typed, semantic, and digest-rebound mutations. A separate stress program checks seven integer probe matrices and 87 exact minimum-energy samples.

For TPC-247, Theorem 2.1 now classifies the vector $(\langle v_{cb}, W_c \rangle)_b$ for each fixed c . It does not estimate the actual Gram matrices, attach primitive rational frequencies, or prove that a longitudinal center dominates any transverse radius. Consequently

ARITHMETIC_L2=NONE, FULL_GATE_B=OPEN, STRICT_1_OVER_400=UNPAID.

The next exact contraction is now forced: combine weighted probes inside each shared output lane, compute the sharp group support radius from its Gram matrix, and only then sum across output blocks.

7 Conclusion

The joint shared-lane geometry is an ellipsoid, not a product of marginal disks. Its range, energy, sphere filling, complex orientation, and grouped budget are all exact, including degenerate ranks and zero radius. This removes TPC-247’s first structural ambiguity while preserving its arithmetic firewall.

Maximum claim. PROVED_STRUCTURAL_L1_SHARED_LANE_GRAM_ELLIPSOID_FEASIBLE_SET.

References

- [1] Liang Wang. Literal v59 source-operator two-lane block attachment, 2026. TPC-247 committed repository paper.
- [2] Liang Wang. Weighted covariance-disk reassembly and hard-window margins, 2026. TPC-246 committed repository paper.